F78PA Probability & Statistics A

Tutorial 9

1. Suppose X and Y are two independent random variables with exponential distribution Exp(1).

   (a) Determine the joint density function $f_{X,Y}(x,y)$ for X and Y and compute $\mathbb{P}\{2X > Y\}$.

   (b) Suppose that
   $$Z = \frac{X}{X+Y}.$$
   Explain why $\mathrm{Im}Z = (0,1)$.
   By integrating over the appropriate region to find $F_Z(z) = \mathbb{P}(Z \leq z)$, show that the random variable Z has a uniform distribution on the interval $(0,1)$.

   (c) Suppose that $W = X + Y$. Show that W has a $\Gamma(2,1)$ distribution.
   Hint: First determine the distribution function $F_W(w)$ for W as in part (b). Then differentiate to obtain the density.

2. Suppose that the random variable X has density $f_X(x) = 2x$ on the interval $(0,1)$ and the random variable Y has the density $f_Y(y) = 3y^2$ on the interval $(0,1)$ and that the variables X and Y are independent. Compute the probability $\mathbb{P}(X + Y \leq 1.4)$.
   Hint: Draw a picture on the unit square, and observe that
   $$\mathbb{P}(X+Y \leq 1.4) = \int_{x=0}^{0.4} \left[\int_{y=0}^{1} f_{X,Y}(x,y)\, dy \right] dx + \int_{x=0.4}^{1} \left[\int_{y=0}^{1.4-x} f_{X,Y}(x,y)\, dy \right] dx$$
   where $f_{X,Y}(x,y)$ is the joint density for X and Y.

3. Suppose that the variables X and Y are jointly continuous and independent. Denoting by f_X and f_Y the marginal density functions for X and Y, respectively, verify that for any $y \in \mathbb{R}$ such that $f_Y(y) > 0$, we have
   $$\mathbb{E}(X \mid Y = y) = \mathbb{E}(X).$$
   Remark This result holds in general – e.g. it also holds for jointly discrete random variables – provided X, Y are **independent**.

4. **Law of Total Expectation (Partition Theorem for Expectation)**

   (a) Suppose that X and Y are jointly continuous with joint density function $f_{X,Y}(x,y)$. Show that
   $$\int_{\mathbb{R}} \mathbb{E}(X \mid Y = y) f_Y(y) dy = \mathbb{E}(X)$$
   where $f_Y(y)$ denotes the marginal density of Y.

   (b) Sam has an alarm clock which goes off each morning at a random time which is uniformly distributed between 8 and 9am. Suppose that X is the time, measured as minutes past 8 am, at which the clock rings (so $0 < X < 60$). Given that $X = x$, the time Y at which Sam gets up, measured as minutes past 8 am, is uniformly distributed between x and 60.

      i. Write down the conditional density $f_{Y|X}(y|x)$ for $0 < x < y < 60$.

      ii. Determine $\mathbb{E}(Y|X = x)$ and hence, or otherwise, compute $\mathbb{E}(Y)$.

5. Suppose that X is a random variable with density $f_X(x) = 3x^2$ on the interval $(0,1)$ and suppose that Y is a random variable with conditional density $f_{Y|X}(y|x) = xe^{-xy}$ when $0 < y < \infty$ and $0 < x < 1$.

(a) Identify the conditional distribution of Y given $X = x$ and explain *without calculation* why for $0 < x < 1$

$$\mathbb{E}(Y \mid X = x) = \frac{1}{x}.$$

(b) Use the Law of Total Expectation to compute $\mathbb{E}(Y)$.

(c) Use the Law of Total Probability for densities to determine $\mathbb{P}\{Y > 3\}$.

(d) Determine the joint density $f_{X,Y}(x, y)$ for the variables X and Y. Hence, or otherwise, determine $\mathbb{P}(Y > X^2)$.

6. Let X and Y be two random variables. Show that

$$\mathbb{V}\text{ar}(X + Y) = \mathbb{V}\text{ar}(X) + \mathbb{V}\text{ar}(Y) + 2\mathbb{C}\text{ov}(X, Y).$$

7. Suppose that X, Y, Z are random variables and a, b, c, d are (non-random) constants. Show that

(a) $\mathbb{C}\text{ov}(a + bX, c + dY) = bd\mathbb{C}\text{ov}(X, Y)$,

(b) $\mathbb{C}\text{ov}(X + Y, Z) = \mathbb{C}\text{ov}(X, Z) + \mathbb{C}\text{ov}(Y, Z)$.

Exam Style Question

Let the joint density function of X and Y be given by

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{y} e^{-\left(y + \frac{x}{y}\right)} & \text{for } 0 < x, y < \infty, \\ 0 & \text{otherwise.} \end{cases}$$

(a) By finding the marginal density, f_Y, show that $Y \sim \text{Exp}(1)$. (3 marks).

(b) (i) For $y > 0$, determine $f_{X|Y}(x|y)$, the conditional density of X given that $Y = y$. (2 marks)

(ii) Explain (without calculation) why, for $y > 0$,

$$\mathbb{E}[X|Y = y] = y.$$

(1 mark)

(iii) Calculate $\mathbb{E}[e^{X/3}|Y = 1]$. (3 marks)

(c) Given that $\mathbb{E}[X] = 1$ and $\mathbb{E}[Y] = 1$, show that $\mathbb{C}\text{ov}(X, Y) = 1$. (2 marks)

(d) Comment on the relationship between the variables X and Y. (1 mark)